EduTrack

Multi-Tier Project Management System for Educational Institutions

Academic Year: 2025-2026
Duration: 3 Months
Platform: Web-based System

Streamlining Academic Project Collaboration

1. Introduction & Objectives

What is EduTrack?

EduTrack is a comprehensive web-based project management system designed specifically for colleges and educational institutions. It enables seamless collaboration between Directors, HODs, Teachers, and Students on academic projects with robust role-based access control.

Project Objectives

- Centralize project assignment, submission, and grading in one platform
- Provide role-based dashboards for Directors, Teachers, and Students
- Enable online code writing, testing, and submission with integrated editor
- Facilitate code review with inline commenting and structured feedback
- Generate real-time performance analytics for all stakeholders
- Reduce administrative overhead by 60% through automation

2. System Requirements Specification (SRS)

Functional Requirements

- User authentication with institutional email
- Role-based access control (RBAC)
- Project creation and assignment
- Online code editor (Monaco)
- File upload/download management
- Code review and grading system
- Performance reports and analytics
- Email notifications

Non-Functional Requirements

- Performance: Response time < 2 seconds
- Scalability: Support 10,000+ users
- Security: JWT auth, encrypted storage
- Availability: 99.9% uptime SLA
- Usability: Responsive design for all devices
- Reliability: Daily automated backups
- Maintainability: Modular architecture
- Compatibility: Chrome, Firefox, Safari, Edge

2. SRS - User Roles & Permissions

Director/HOD

- View all teachers and students
- Access complete system reports
- Manage teacher accounts
- Oversee multiple departments
- Generate analytics (PDF/Excel)

Teacher

- Create and assign projects
- Bulk import students (CSV)
- Review code with inline comments
- Grade submissions with rubrics
- View student performance reports

Student

- View assigned projects only
- Write code in online editor
- Upload files and screenshots
- Submit multiple revisions
- Track personal performance

3. Process Logic - System Workflow

1	Director/HOD creates teacher accounts	Admin
2	Teacher bulk imports students via CSV	Teacher
3	Teacher creates and assigns project	Teacher
4	Student receives notification via email	Student
5	Student works on code in online editor	Student
6	Student submits code and files	Student
7	Teacher reviews code with inline comments	Teacher
8	Teacher grades using structured rubric	Teacher
9	Student views grade and feedback	Student
10	Director generates department analytics	Admin

4. Gantt Chart - Project Timeline (3 Months)

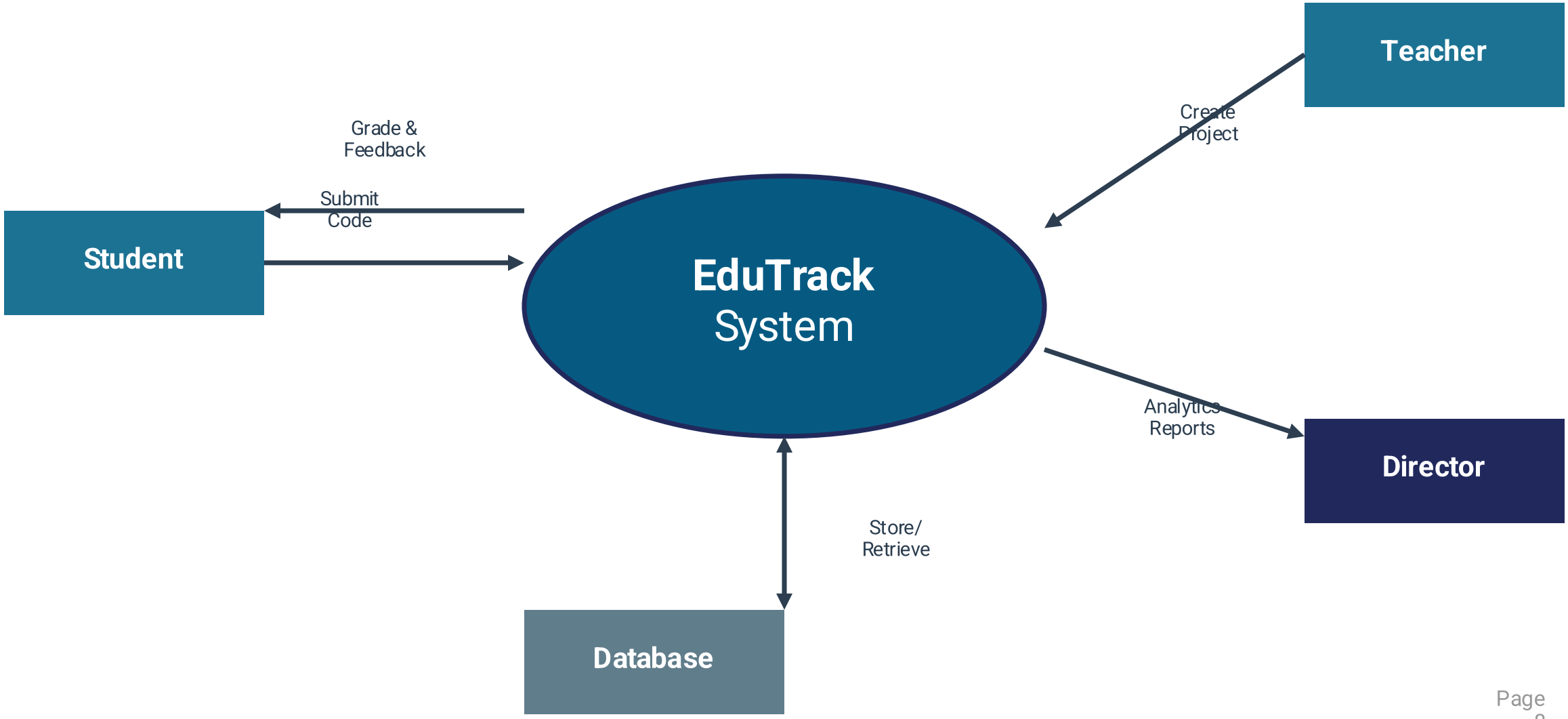
Phase	Month 1	Month 2	Month 3
Requirements & Design	✓ Complete		
Database & Backend	✓ Complete	✓ Complete	
Frontend Development		In Progress	Planned
Code Editor Integration		In Progress	
Testing & QA			Planned
Deployment & Training			Planned

Status Legend: Complete · In Progress · Planned

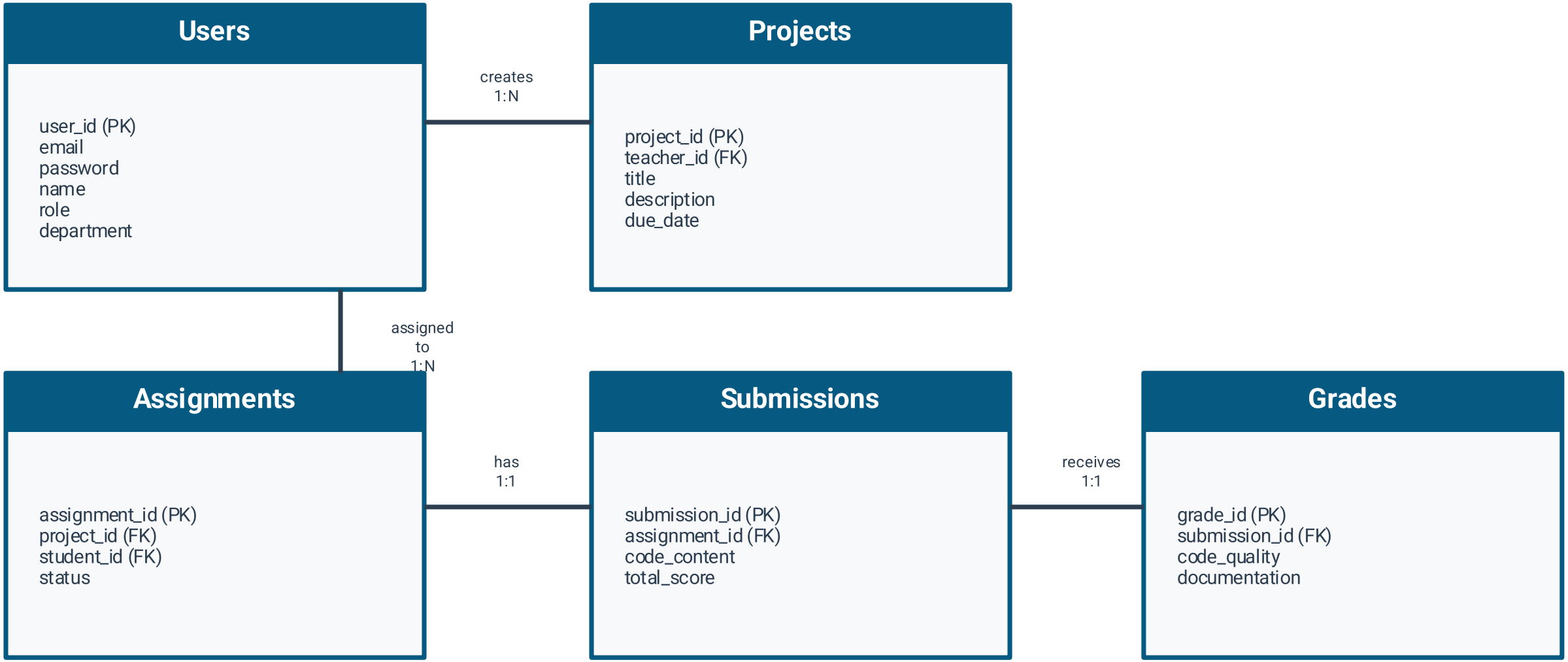
5. Data Dictionary

Entity	Attribute	Data Type	Description
Users	user_id	INT (PK)	Unique identifier for user
	email	VARCHAR(255)	Institutional email address
	password	VARCHAR(255)	bcrypt hashed password
	role	ENUM	student, teacher, director
	department	VARCHAR(50)	Academic department
Projects	project_id	INT (PK)	Unique identifier for project
	teacher_id	INT (FK)	Reference to users table
	title	VARCHAR(200)	Project title
	due_date	TIMESTAMP	Submission deadline
	max_marks	INT	Maximum score (default: 100)
Assignments	assignment_id	INT (PK)	Unique identifier
	project_id	INT (FK)	Reference to projects
	student_id	INT (FK)	Reference to users
	status	ENUM	assigned, submitted, graded
Submissions	submission_id	INT (PK)	Unique identifier
	assignment_id	INT (FK)	Reference to assignments
	code_content	TEXT	Student's code
	file_urls	JSONB	Array of S3 file URLs
	total_score	INT	Final grade (0-100)

6. Data Flow Diagram (Level 0 - Context Diagram)



7. Entity-Relationship Diagram



PK = Primary Key, FK = Foreign Key

8. System Interfaces - User Dashboards

Student Dashboard

📋 Active Tasks (5)
✅ Submitted (3)
⭐ Graded (2)
📊 Performance: 82% avg
💻 Code Editor Access
📁 File Upload (PDF, ZIP, PNG)
🔔 Notifications

Teacher Dashboard

📝 Create New Project
📊 My Projects (8 active)
⏳ Pending Reviews (15)
✍️ Code Review Interface
📈 Student Analytics
📤 Bulk Import Students
⚙️ Grading Rubrics

8. Code Editor Interface (Monaco Editor)

Main.java

```
public class BankAccount {
    private double
balance;

    public void
deposit(double amt) {
        balance += amt;

    }

}
```

[▶ Run Code] [💾 Save] [📤 Submit]

Editor Features:

✓ Syntax Highlighting ✓ Auto-completion ✓ Error Detection ✓ Auto-save (30s) ✓ Multi-language Support ✓ Code Execution

9. Expected Report Generation

Student Performance Report

- Total projects assigned: 12
- Completed on time: 10 (83%)
- Average grade: 85/100
- Strengths: Code quality, documentation
- Areas for improvement: Error handling

Teacher Workload Report

- Active projects: 8
- Students managed: 120
- Pending reviews: 15
- Average grading time: 12 mins
- Completion rate: 88%

Department Analytics

- Total students: 450
- Active projects: 35
- Department average: 78%
- Top performing stream: BCA
- Projects submitted: 85%

Export Formats Available:

📄 PDF Reports - Print-ready format
📊 Excel Spreadsheets - Data analysis
📈 Visual Charts - Performance graphs
📧 Email Delivery - Automated distribution

Technology Stack

Layer	Technology	Purpose
Frontend	React.js, Tailwind CSS, Monaco Editor	User interface and code editing
Backend	Node.js, Express.js, JWT Authentication	RESTful API and business logic
Database	PostgreSQL, Redis Cache	Data persistence and caching
Storage	AWS S3	File and code submission storage
Code Execution	Judge0 API, Docker Containers	Sandboxed code running
Email	SendGrid SMTP Service	Notifications and alerts
Deployment	AWS EC2, CloudFront CDN, Route 53	Hosting and content delivery

10. References & Bibliography

Technical Documentation:

- React.js Official Documentation - https://react.dev
- Node.js Documentation - https://nodejs.org/docs
- PostgreSQL 15 Documentation - https://www.postgresql.org/docs
- Monaco Editor API Reference - https://microsoft.github.io/monaco-editor
- AWS S3 Developer Guide - https://aws.amazon.com/s3/getting-started
- JWT Authentication Best Practices - https://jwt.io/introduction

Academic References:

- Software Engineering: A Practitioner's Approach - Roger Pressman (9th Edition)
- Database System Concepts - Silberschatz, Korth, Sudarshan
- Web Application Architecture - Leon Shklar & Rich Rosen
- Design Patterns for Cloud-Native Applications - Kasun Indrasiri
- IEEE Standards for Software Requirements Specification (IEEE 830)

11. Future Scope & Enhancements

AI-Powered Features

- Auto code review suggestions
- Plagiarism detection using ML
- Intelligent grading assistance
- Chatbot for student support

Mobile Applications

- iOS app (Swift/SwiftUI)
- Android app (Kotlin)
- Push notifications
- Offline mode support

Integrations

- LMS integration (Moodle, Canvas)
- GitHub repository sync
- Google Workspace integration
- Microsoft Teams connector

Advanced Analytics

- Predictive performance insights
- Learning pattern analysis
- Skill gap identification
- Personalized recommendations

Collaboration Tools

- Peer code review system
- Group project management
- Video conferencing integration
- Real-time collaboration

Enhanced Security

- Two-factor authentication
- Biometric login (mobile)
- Advanced audit logging
- GDPR compliance features

Thank You

Questions & Discussion

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Demo: https://demo.edutrack.edu

Transforming Academic Project Management